FN2 Worked Examples: Function Notation

Recording companion for the FN2 CheckIt support video — Math 172, Santa Ana College

How to use this document

- Each worked example is on its own page, in the same order as the Try It tabs in the interactive explainer, with the same numbers.
- The right-hand third of every page is intentionally blank. It is space for handwriting or ink notation while you narrate (for example, after inserting this PDF as a printout in OneNote).
- Say it at the start of the video: your CheckIt problem will have different numbers than the examples here. The reasoning is what transfers.

The one question behind every example

Am I given the input and finding the output, or given the output and finding the inputs? f(a) means the output when the input is a: replace x with a, using the piece whose condition a satisfies. g(x) = b gives the output and asks for every input that produces it: on a graph, draw the horizontal line y = b and read the x-value at each crossing.

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Example 1: At the Cut

$$f(x)=\begin{cases} 2x^3-3x, & \text{if } -5<x\le 1 \\ \sqrt{(3x^2+5)}, & \text{if } 1<x<5 \end{cases}$$

Question

Find f(1).

A. $2\sqrt{2}$
B. 5
C. −1
D. Undefined, since 1 is where the pieces meet

Answer and reasoning

The condition $-5<x\le 1$ includes 1, because of the "or equal," so use $2x^3-3x$: $2(1)^3-3(1)=2-3=-1$. The other piece, $1<x<5$, leaves 1 out. $2\sqrt{2}=\sqrt{8}$ uses the wrong piece; 5 adds the $3x$ instead of subtracting; and the function is defined at the cut, since exactly one piece includes it. ***f(1)=−1*.**

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Example 2: Left of the Cut

$$f(x)=\begin{cases} 2x^3-3x, & \text{if } -5<x\le 1 \\ \sqrt{(3x^2+5)}, & \text{if } 1<x<5 \end{cases}$$

Question

Find f(−2).

A. −10
B. 22
C. −22
D. $\sqrt{17}$

Answer and reasoning

−2 is between −5 and 1, so use $2x^3-3x$: $2(-2)^3-3(-2)=2(-8)+6=-16+6=-10$. 22 uses $(-2)^3=8$, but a negative cubed stays negative; −22 subtracts the 6 instead of adding it, since $-3(-2)=+6$; and $\sqrt{17}$ uses the radical piece, which only covers $1<x<5$. ***f(−2)=−10*.**

Example 3: Right of the Cut

$$f(x)=\begin{cases} 2x^3-3x, & \text{if } -5<x\le 1 \\ \sqrt{(3x^2+5)}, & \text{if } 1<x<5 \end{cases}$$

Question

Find f(3). Give an exact, simplified answer.

A. $\sqrt{14}$
B. 45
C. $\sqrt{86}$
D. $4\sqrt{2}$

Answer and reasoning

3 is between 1 and 5, so use the radical: $\sqrt{(3(3)^2+5)}=\sqrt{(27+5)}=\sqrt{32}=\sqrt{(16\cdot 2)}=4\sqrt{2}$. $\sqrt{14}$ multiplies 3 by 3 instead of by 3^2; $\sqrt{86}$ squares $3\cdot 3$ instead of only the x; and 45 uses the polynomial piece. ***f(3)=4√2*.**

Example 4: A Different Cut

$$k(x)=\begin{cases}(4x-3)/(2x^2+1), & \text{if } -5<x<-1\\ x^2+2x, & \text{if } -1\le x<5\end{cases}$$

Question

Find k(−1).

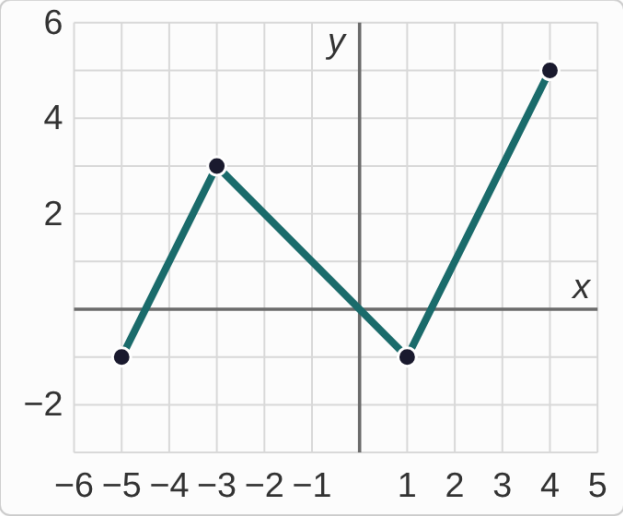
A. $-7/3$
B. −1
C. 3
D. Undefined, since −1 is where the pieces meet

Answer and reasoning

This time the "or equal" is on the right-hand piece: $-1\le x<5$ includes −1. So $k(-1)=(-1)^2+2(-1)=1-2=-1$. $-7/3$ uses the fraction piece, whose condition $x<-1$ leaves −1 out; 3 treats $2(-1)$ as $+2$; and the function is defined at the cut. ***k(−1)=−1*.**

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Example 5: Input to Output 1



Plotted data: A graph of g on the interval from x = −5 to x = 4, drawn on a grid with lines every 1 unit. It is made of three straight segments: from (−5, −1) up to (−3, 3), then down to (1, −1), then up to (4, 5). The ends and corners are marked with dots.

Question

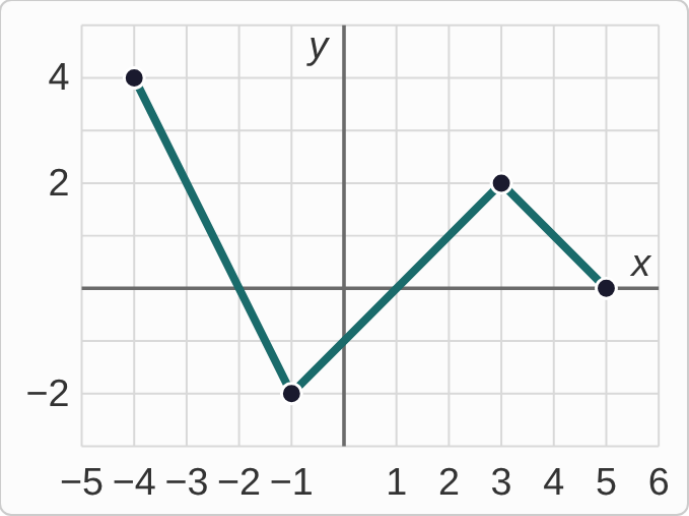
Find g(−1).

A. −1
B. 0
C. 3
D. 1

Answer and reasoning

Start at $x=-1$ on the x-axis and go straight up to the graph. You land on the segment from $(-3,3)$ to $(1,-1)$, which drops 1 for every step right, so at $x=-1$ it is at height $3-2=1$. −1 is the height at $x=1$, not $x=-1$; 0 is the height at $x=0$; and 3 is the peak at $x=-3$. ***g(−1)=1*.**

Example 6: Input to Output 2



Plotted data: A graph of g on the interval from x = −4 to x = 5, drawn on a grid with lines every 1 unit. It is made of three straight segments: from (−4, 4) down to (−1, −2), then up to (3, 2), then down to (5, 0). The ends and corners are marked with dots.

Question

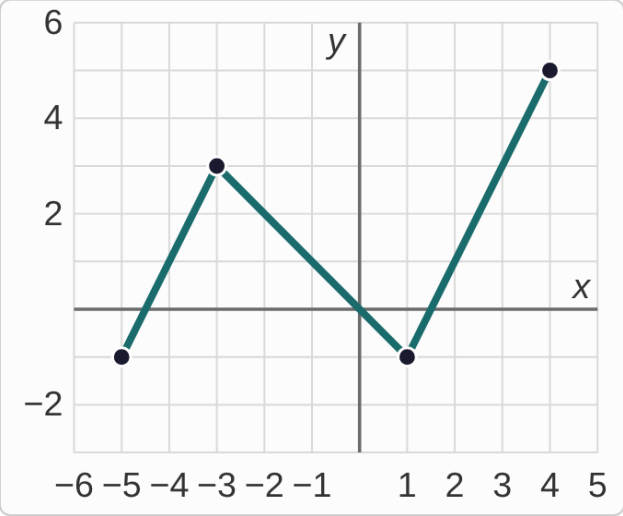
Find g(3).

A. 3
B. 2
C. 1
D. −2

Answer and reasoning

At $x=3$ the graph has a corner at $(3,2)$, so $g(3)=2$. A corner belongs to both segments, and they agree there. 3 repeats the input instead of reading the output; 1 is the height at $x=2$; and −2 is the lowest point, at $x=-1$. ***g(3)=2*.**

Example 7: Output to Inputs 1



Plotted data: A graph of g on the interval from x = −5 to x = 4, drawn on a grid with lines every 1 unit. It is made of three straight segments: from (−5, −1) up to (−3, 3), then down to (1, −1), then up to (4, 5). The ends and corners are marked with dots.

Question

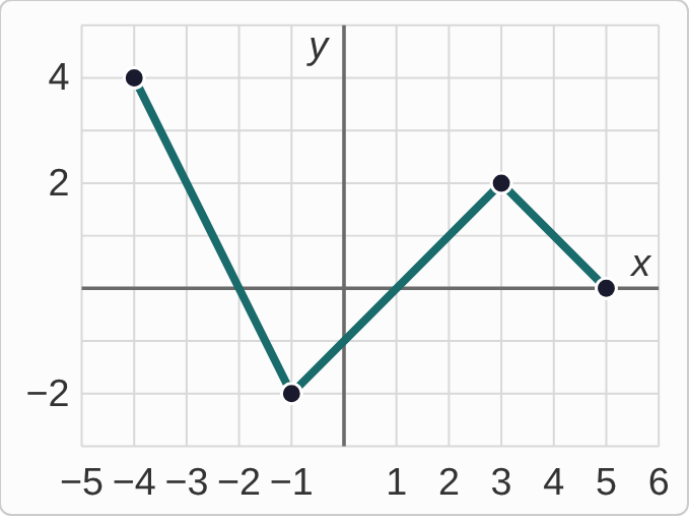
For which x-values in [−5, 4] does g(x) = 2?

- **A.** x = −3.5, x = −2, and x = 2.5
- **B.** x = −2 only, the crossing at a whole number
- **C.** x = −3.5 and x = 2.5
- **D.** x = −4, x = −2, and x = 2

Answer and reasoning

Draw the horizontal line $y=2$. It crosses all three segments. First segment, rising 2 per step from $(-5,-1)$: it reaches 2 after 1.5 steps, at $x=-3.5$. Second, falling 1 per step from $(-3,3)$: 1 step, at $x=-2$. Third, rising 2 per step from $(1,-1)$: 1.5 steps, at $x=2.5$. "−2 only" keeps just the crossing at a whole number; leaving out −2 misses the middle segment; and −4 and 2 round the halves down. ***x*=−3.5, *x*=−2, and *x*=2.5.**

Example 8: Output to Inputs 2



Plotted data: A graph of g on the interval from x = −4 to x = 5, drawn on a grid with lines every 1 unit. It is made of three straight segments: from (−4, 4) down to (−1, −2), then up to (3, 2), then down to (5, 0). The ends and corners are marked with dots.

Question

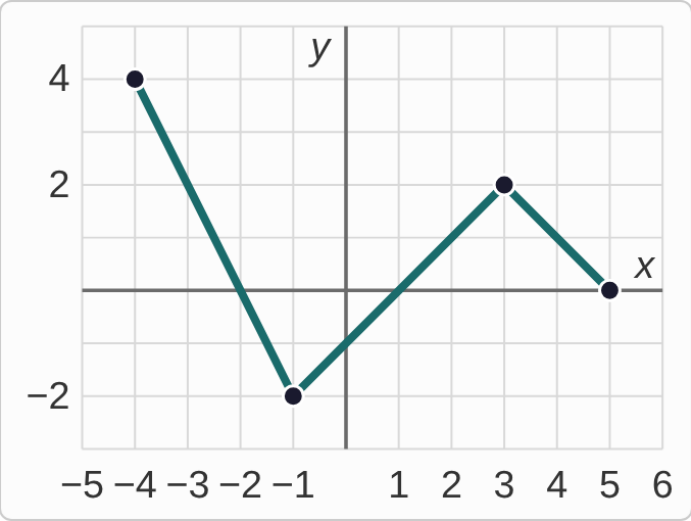
For which x-values in [−4, 5] does g(x) = 0?

- **A.** x = −2 and x = 1, the two crossings
- **B.** x = 1 only
- **C.** x = −2, x = 1, and x = 5
- **D.** x = −1, since g(0) = −1

Answer and reasoning

g(x)=0 asks where the graph is on the *x*-axis. It crosses at *x=−2*, falling 2 per step from *(−4,4)*; at *x=1*, rising from *(−1,−2)*; and it ends on the axis at *(5,0)*, which counts, since 5 is in the interval. Stopping at two misses the endpoint; "1 only" misses the first segment; and *g(0)=−1* answers a different question, the output at input 0. ***x=−2*, *x=1*, and *x=5*.**

Example 9: Just One



Plotted data: A graph of g on the interval from x = −4 to x = 5, drawn on a grid with lines every 1 unit. It is made of three straight segments: from (−4, 4) down to (−1, −2), then up to (3, 2), then down to (5, 0). The ends and corners are marked with dots.

Question

For which x-values in [−4, 5] does g(x) = −2?

- **A.** x = −2
- **B.** x = −1 and x = 3, the two corners
- **C.** There is none, since corners do not count.
- **D.** x = −1 only

Answer and reasoning

Draw the horizontal line $y=-2$. It touches the graph only at the lowest corner, $(-1,-2)$, so there is exactly one input: $x=-1$. Many CheckIt problems have just one answer like this. $x=-2$ copies the output as an input; the corner at 3 is at height 2, not −2; and a corner is a point on the graph like any other. ***x=−1* only.**

Example 10: What It Means

$$C(10)=150$$

Question

C(m) is the temperature of a cup of coffee, in degrees Fahrenheit, m minutes after it is poured. What does C(10) = 150 mean?

A. 10 minutes after it is poured, the coffee is 150°F.
B. 150 minutes after it is poured, the coffee is 10°F.
C. The coffee cools by 150 degrees in 10 minutes.
D. C times 10 equals 150, so C is 15.

Answer and reasoning

The input, in the parentheses, is the time: 10 minutes. The output is the temperature: 150°F. Swapping them reads the notation backwards; "cools by 150" turns an output into a change, which the notation does not say; and $C(10)$ is not multiplication: C is the name of the function. **10 minutes after it is poured, the coffee is 150°F.**

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Example 11: On the Graph

$$h(-2)=5$$

Question

Which point must be on the graph of h?

- **A.** (5, −2)
- **B.** (−2, −5)
- **C.** (0, 5)
- **D.** (−2, 5)

Answer and reasoning

$h(-2)=5$ says the input −2 gives the output 5, and points are written (input, output). So $(-2,5)$. $(5,-2)$ swaps input and output; $(-2,-5)$ changes the output's sign; and $(0,5)$ would mean $h(0)=5$. ***(−2,5).***

Example 12: Back to the Input

$$f(x)=3x-4$$

Question

For which x does f(x) = 11?

A. x = 29
B. x = 5
C. x = 7/3
D. x = 15

Answer and reasoning

Set the output equal to 11 and solve for the input: $3x-4=11$, $3x=15$, $x=5$. Check: $f(5)=15-4=11$. 29 is $f(11)$, the output for input 11, the opposite question; $7/3$ subtracts the 4 instead of adding it; and 15 stops before dividing by 3. ***x=5*.**

Example 13: Slip 1

$$m(x)=\begin{cases} 3x^2+1, & \text{if } -5<x<0 \\ \sqrt{(x^2+5)}, & \text{if } 0\le x<5 \end{cases}$$

Question

A student found m(0) = 3(0)² + 1 = 1. What is the mistake, if any?

A. There is no mistake.
B. 0 belongs to the piece with 0 ≤ x, so m(0) = √5.
C. m(0) is undefined, since 0 is where the pieces meet.
D. 0² is 1, so 3(0)² + 1 is 4.

Answer and reasoning

$-5<x<0$ leaves 0 out, and $0\le x<5$ includes it, so $m(0)=\sqrt{(0^2+5)}=\sqrt{5}$. The function is defined at 0, since one piece includes it, and $0^2=0$, not 1, so $3(0)^2+1$ really is 1, just from the wrong piece. **0 belongs to the radical piece: $m(0)=\sqrt{5}$.**

Example 14: Slip 2

$$m(-2)=3(-2)^2+1$$
$$=-12+1$$
$$=-11$$

Question

Using the same m, a student found m(−2) this way. What is the mistake, if any?

A. There is no mistake.
B. −2 belongs to the radical piece, so m(−2) = 3.
C. It should be 3(−2) + 1 = −5, without the square.
D. $(-2)^2$ is 4, not −4, so m(−2) = 13.

Answer and reasoning

The piece is right, since −2 is between −5 and 0. But $(-2)^2=(-2)(-2)=4$, so $m(-2)=3(4)+1=13$. The radical piece starts at 0, and the square is part of the formula; dropping it changes the function. ***$(-2)^2=4$*: *$m(-2)=13$*.**

Example 15: Slip 3

$$p(x)=\begin{cases}(4x+2)/(2x^2+2), & \text{if } -5<x<-1\\ x^2+2x, & \text{if } -1\le x<5\end{cases}$$

Question

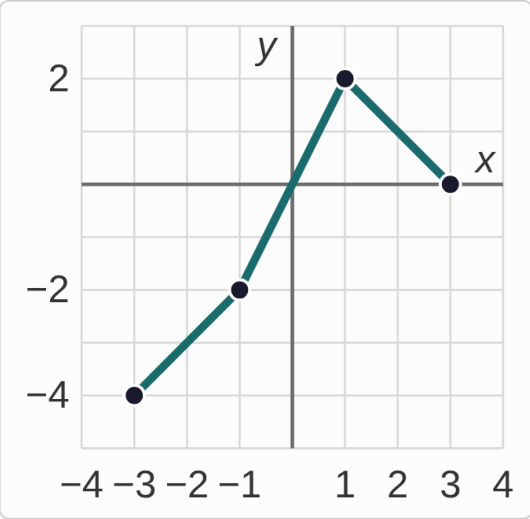
A student found $p(-2)=(4(-2)+2)/(2(-2)^2+2)=(-6)/10=-3/5$. What is the mistake, if any?

A. There is no mistake.
B. −2 belongs to the other piece, so p(−2) = 0.
C. The bottom should be 2(−4) + 2 = −6, so it is 1.
D. It should stay −6/10, because reducing changes the value.

Answer and reasoning

−2 is between −5 and −1, so the fraction piece is the right one. Top: $4(-2)+2=-6$. Bottom: $2(-2)^2+2=2(4)+2=10$. So $p(-2)=-6/10=-3/5$, reduced, as CheckIt wants. The other piece starts at −1; $(-2)^2$ is 4, not −4; and reducing a fraction never changes its value. **There is no mistake.**

Example 16: Slip 4



Plotted data: A graph of g on the interval from x = −3 to x = 3, drawn on a grid with lines every 1 unit. It is made of three straight segments: from (−3, −4) up to (−1, −2), then up more steeply to (1, 2), then down to (3, 0). The ends and corners are marked with dots.

Question

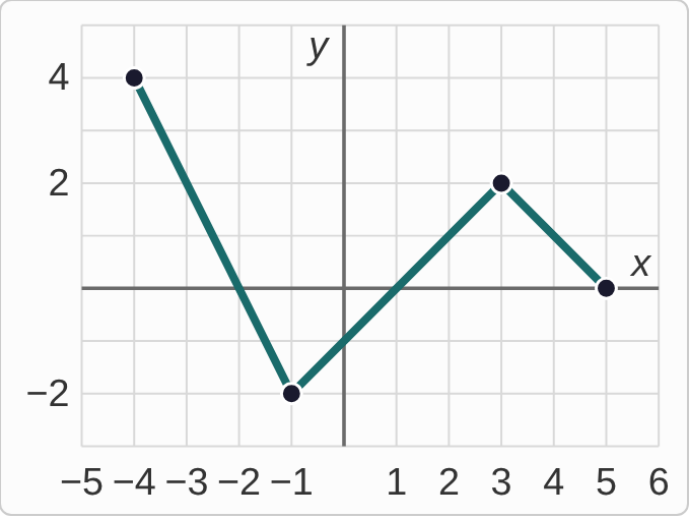
For which x in [−3, 3] does g(x) = 0? A student answered x = 0 only. What is the mistake, if any?

- **A.** There is no mistake.
- **B.** x = 3 counts too: the right end is on the x-axis.
- **C.** The answer should be the output g(0) = 0, not an x-value.
- **D.** x = −1 counts too, since the graph turns there.

Answer and reasoning

The line $y=0$ is the x-axis. The steep segment from $(-1,-2)$ to $(1,2)$ crosses it at $x=0$, and the graph ends on it at $(3,0)$. The end is a point of the graph, and 3 is in the interval, so $x=0$ and $x=3$. The question asks for inputs, so the answer is x-values; and at $x=-1$ the graph is at height −2, where it gets steeper but does not turn around. ***x=0* and *x=3*.**

Example 17: Slip 5



Plotted data: A graph of g on the interval from x = −4 to x = 5, drawn on a grid with lines every 1 unit. It is made of three straight segments: from (−4, 4) down to (−1, −2), then up to (3, 2), then down to (5, 0). The ends and corners are marked with dots.

Question

A student was asked for g(4) and answered −4, because the graph is at height 4 when x = −4. What is the mistake, if any?

- **A.** There is no mistake.
- **B.** g(4) is 0, since the graph ends on the x-axis at the right end.
- **C.** Only the sign is off: the height is 4, so g(4) = 4.
- **D.** That solves g(x) = 4. At x = 4, the height is 1.

Answer and reasoning

g(4) gives the input 4 and asks for the output: go to *x=4* and read the height. The segment from *(3,2)* to *(5,0)* drops 1 per step, so at *x=4* it is at 1. The student answered the reverse question, *g(x)=4*. 0 is the height at the right end, *x=5*, not at *x=4*; and 4 is the output the student matched, not the output at input 4. ***g(4)=1*.**

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